Numerical Methods

Roots of Non-linear equations

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Roots of Non-linear equations

- Roots of polynomials
  - Polynomials
  - roots
- Roots of systems of non-linear equations
  - Fixed-point iteration
  - Newton-Raphson

- Polynomials: the general form

$$f_n(x) = a_n x^n + a_{n-1} x^{n-1} + \ldots + a_1 x + a_0$$

or

$$f_n(x) = \sum_{i=0}^{n} a_i x^i$$

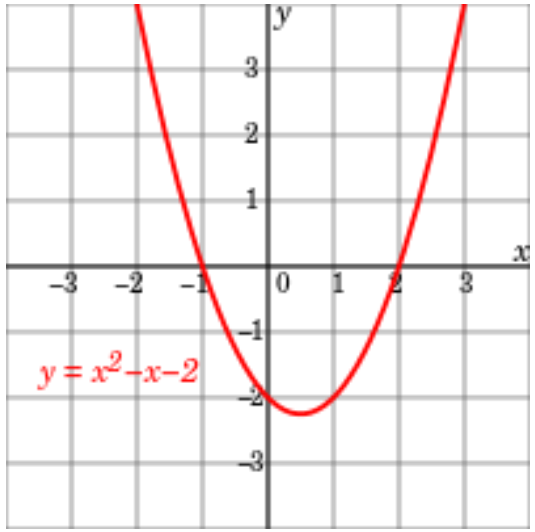
- where *n* is the order of the polynomial, and the *a's* are constant coefficients
- Polynomials have many applications in engineering and science.
  - They are used extensively in curve-fitting.
  - One of their most interesting and powerful applications is in characterizing dynamic systems and, in particular, linear systems. Examples include mechanical devices, structures, and electrical circuits.

- Polynomials are a special type of nonlinear algebraic equation

$$f_n(x) = a_n x^n + a_{n-1} x^{n-1} + \ldots + a_1 x + a_0 = 0 \quad \text{or} \quad f_n(x) = 0$$

- The solutions of this equation are called the roots of the polynomial, In general, an *nth* order polynomial will have *n* roots (can be real and/or complex)
- If n is odd, there is at least one real root.
- they are the zeroes of the function f (corresponding to the points where the graph of f meets the x-axis)

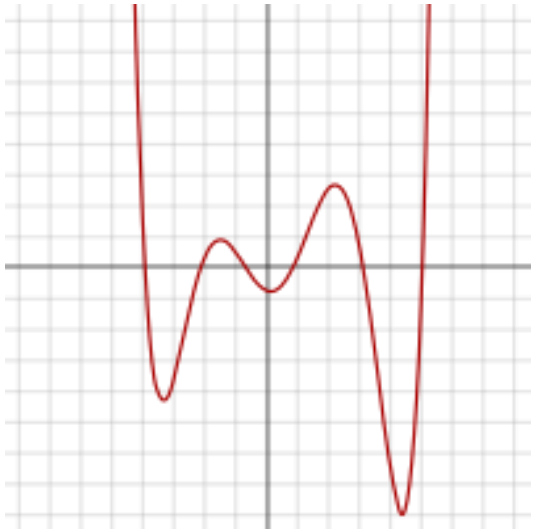
- A polynomial function in one real variable can be represented by a graph



The graph of a polynomial function of degree 2



The graph of a polynomial function of degree 3



The graph of a polynomial function of degree 6

Characteristics of polynomials

- The relation between the roots of a polynomial and its coefficients is described by Vieta's formulas

$$f(x) = ax^2 + bx + c$$

Roots x_1, x_2 of the equation $f(x) = 0$ satisfy

$$x_1 + x_2 = -\frac{b}{a}\ ,\ x_1 x_2 = \frac{c}{a}$$

- In general case, Vieta's formulas

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

$$\begin{cases} x_1 + x_2 + \cdots + x_{n-1} + x_n = -\dfrac{a_{n-1}}{a_n} \\ (x_1x_2 + x_1x_3 + \cdots + x_1x_n) + (x_2x_3 + x_2x_4 + \cdots + x_2x_n) + \cdots + x_{n-1}x_n = \dfrac{a_{n-2}}{a_n} \\ \quad \vdots \\ x_1x_2 \ldots x_n = (-1)^n \dfrac{a_0}{a_n}. \end{cases}$$

- To find the roots of any given monic polynomial, two problems need to be solved:
    1. construct a square matrix A whose characteristic polynomial
    2. find the eigenvalues of this matrix A

Companion matrix

In general, the monic polynomial of degree n

$$p(x) = x^n + a_1 x^{n-1} + \cdots + a_{n-1} x + a_n$$

A companion matrix

$$A = \begin{pmatrix} -a_1 & -a_2 & \cdots & -a_{n-1} & -a_n \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & \\ \cdot & \cdot & \cdots & \cdot & \cdot \\ 0 & 0 & \cdots & 1 & 0 \end{pmatrix}$$

An n-by-n matrix (square matrix)

Characteristics: x_i is a root of the polynomial $p(x)$

$\Rightarrow$ x_i is also a eigenvalue of the companion matrix A

Example

- Suppose we have a polynomial

$$a_1x^5 + a_2x^4 + a_3x^3 + a_4x^2 + a_5x + a_6 = 0$$

Dividing by a_1 and rearranging yields

$$x^5 = -\frac{a_2}{a_1}x^4 - \frac{a_3}{a_1}x^3 - \frac{a_4}{a_1}x^2 - \frac{a_5}{a_1}x - \frac{a_6}{a_1}$$

A special matrix (5x5)

polynomial's companion matrix

$$\begin{bmatrix} -a_2/a_1 & -a_3/a_1 & -a_4/a_1 & -a_5/a_1 & -a_6/a_1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

Calculate the eigenvalues?

Example

- Suppose we have a polynomial $f(x) = x^3 - 7x + 6$

polynomial's companion matrix

$$A = \begin{bmatrix} 0 & 7 & -6 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

Eigenvalue

- The eigenvalues of matrix A are values of λ that satisfy the equation

$$|A - \lambda I| = 0$$

- The fundamental theorem of algebra implies that the characteristic polynomial of an *n* by *n* matrix A, being a polynomial of degree *n*, can be factored into the product of *n* linear terms

$$|A - \lambda I| = (\lambda_1 - \lambda)(\lambda_2 - \lambda)\cdots(\lambda_n - \lambda) = 0$$ ➡ Roots

Eigenvalues : characteristic roots, characteristic values, proper values

Case study 1

Find the eigenvalue of a matrix $A = \begin{bmatrix} 3 & 6 & -8 \\ 0 & 0 & 6 \\ 0 & 0 & 2 \end{bmatrix}$

Step 1: We need to solve the equation

Step 2: Calculation

Step 3: calculate the determinant: det

- Example

Find the eigenvalue of a matrix

$$A = \begin{pmatrix} 1 & 1 \\ -2 & 4 \end{pmatrix}$$

x1 = 2, x2 = 3

Case study 2

Find the roots of a polynomial $f(x) = x^3 - 7x + 6$

Step 1: the companion matrix

Step 2: The eigenvalues of A

Step 3: The eigenvalues of A are the polynomial roots

- Example

Find the roots of a polynomial

$$f(x) = x^3 - 13x - 12$$

X1 = 4, x2 = -1, x3 = -3

- Roots of polynomials
  - Bisection and Newton-Raphson method for real roots
  - Müller and Bairstow methods for real/complex roots (p.181)

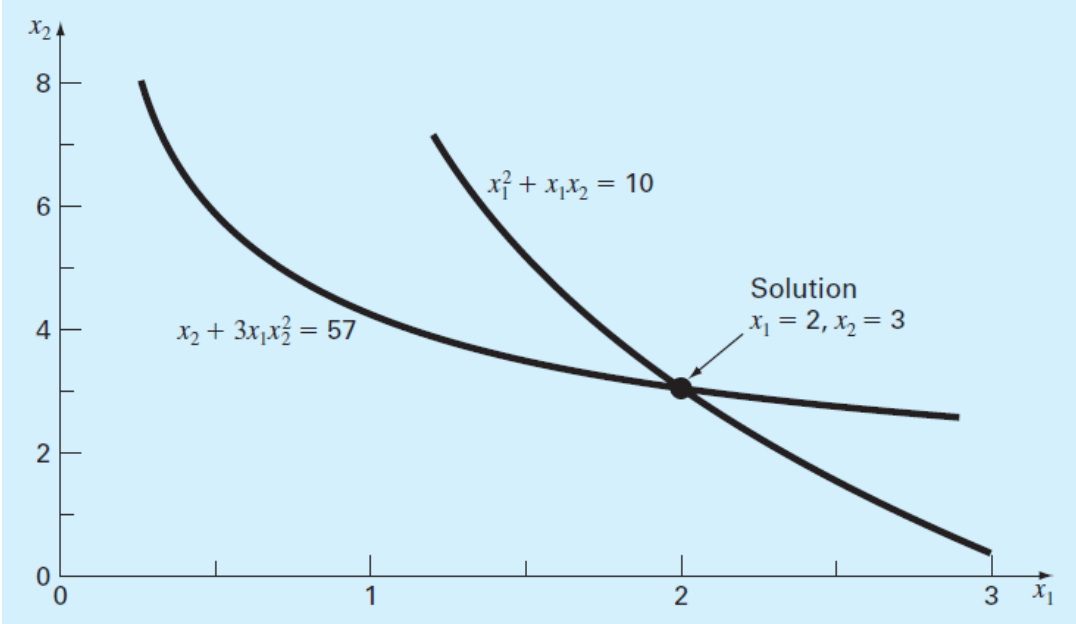
- Roots of polynomials
  - Polynomials
  - roots
- Systems of non-linear equations
  - Fixed-point iteration
  - Newton-Raphson

- Objectives: the primary objective is to find the root of a system of nonlinear equation

$$f_1(x_1, x_2, \ldots, x_n) = 0$$
$$f_2(x_1, x_2, \ldots, x_n) = 0$$
$$\vdots \qquad \vdots$$
$$f_n(x_1, x_2, \ldots, x_n) = 0$$

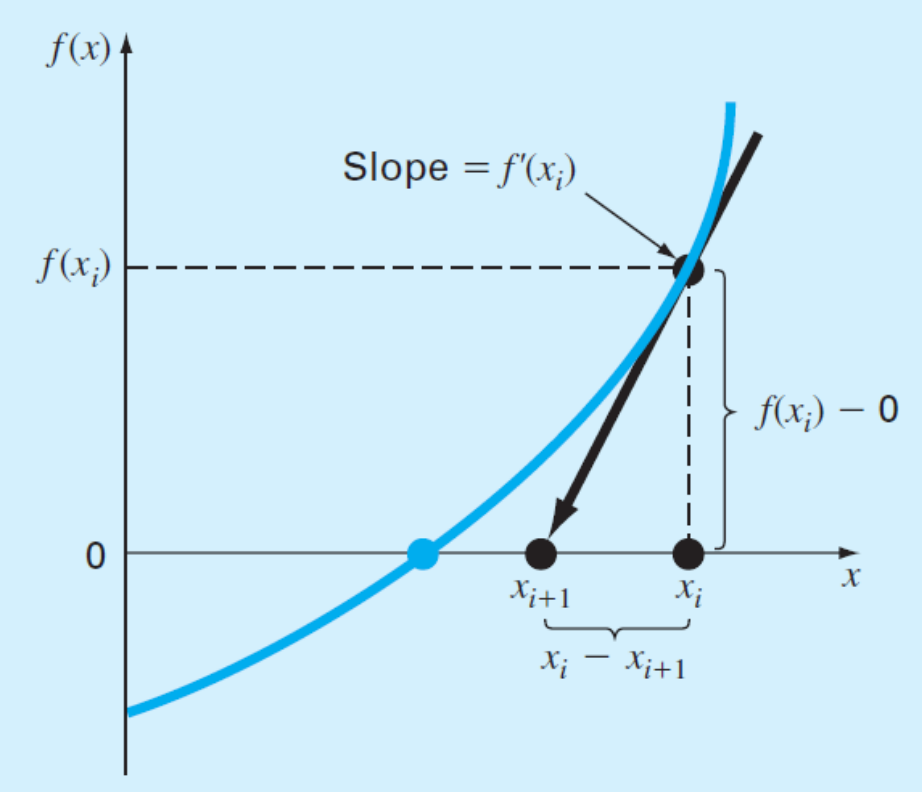
Example

$$x_1^2 + x_1 x_2 = 10$$
$$x_2 + 3x_1 x_2^2 = 57$$

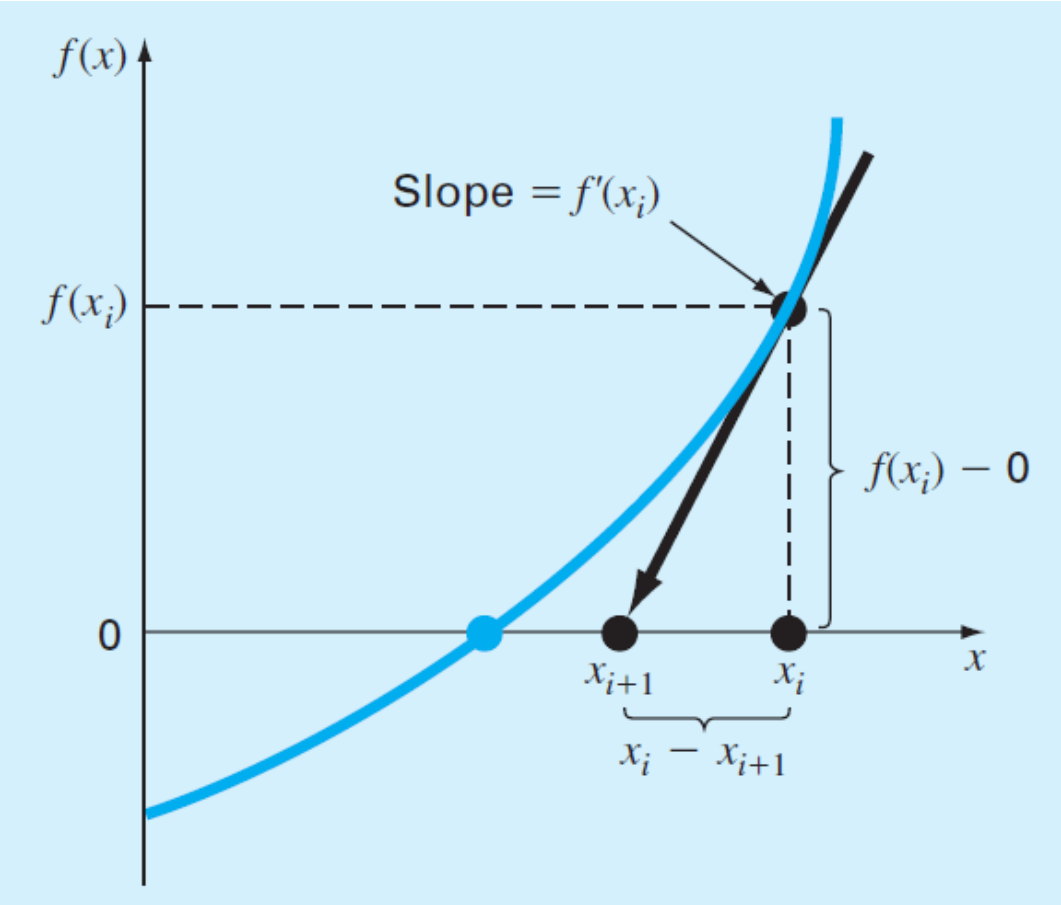


the solution is the intersection of the curves

- Recall that the Newton-Raphson method was predicated on employing the derivative (that is, the slope) of a function to estimate its intercept with the axis of the independent variable—that is, the root.



Newton-Raphson



$$f(x) = 0$$

the first derivative at x is equivalent to the slope

$$f'(x_i) = \frac{f(x_i) - 0}{x_i - x_{i+1}}$$

which can be rearranged to yield

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

Single-equation forme Newton-Raphson

⇨ Systems of non-linear equations

- This estimate was based on a first-order Taylor series expansion

$$f(x_{i+1}) = f(x_i) + (x_{i+1} - x_i) f'(x_i) \quad \Rightarrow \quad f(x_{i+1}) = 0$$

x_{i+1} is solution

$$\Rightarrow \quad x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

which is the single-equation form of the Newton-Raphson method.
=> The multi-equation form is derived in an identical fashion. However, a multivariable Taylor series must be used

- Newton-Raphson

the two-equation case

$$\begin{cases} u(x,y)=0 \\ v(x,y)=0 \end{cases}$$

$$u_{i+1} = u_i + (x_{i+1} - x_i)\frac{\partial u_i}{\partial x} + (y_{i+1} - y_i)\frac{\partial u_i}{\partial y}$$

$$v_{i+1} = v_i + (x_{i+1} - x_i)\frac{\partial v_i}{\partial x} + (y_{i+1} - y_i)\frac{\partial v_i}{\partial y}$$

a first-order Taylor series

the two-equation Newton-Raphson

$$u_{i+1} = 0$$

$$v_{i+1} = 0$$

$$\Rightarrow$$

$$\frac{\partial u_i}{\partial x}x_{i+1} + \frac{\partial u_i}{\partial y}y_{i+1} = -u_i + x_i\frac{\partial u_i}{\partial x} + y_i\frac{\partial u_i}{\partial y}$$

$$\frac{\partial v_i}{\partial x}x_{i+1} + \frac{\partial v_i}{\partial y}y_{i+1} = -v_i + x_i\frac{\partial v_i}{\partial x} + y_i\frac{\partial v_i}{\partial y}$$

- Newton-Raphson

$$\begin{cases} u(x,y)=0 \\ v(x,y)=0 \end{cases}$$

the two-equation Newton-Raphson

$$\Rightarrow \quad x_{i+1} = x_i - \frac{u_i \frac{\partial v_i}{\partial y} - v_i \frac{\partial u_i}{\partial y}}{\frac{\partial u_i}{\partial x}\frac{\partial v_i}{\partial y} - \frac{\partial u_i}{\partial y}\frac{\partial v_i}{\partial x}} \qquad y_{i+1} = y_i - \frac{v_i \frac{\partial u_i}{\partial x} - u_i \frac{\partial v_i}{\partial x}}{\frac{\partial u_i}{\partial x}\frac{\partial v_i}{\partial y} - \frac{\partial u_i}{\partial y}\frac{\partial v_i}{\partial x}}$$

The two-equation Newton-Raphson approach can be generalized to solve n simultaneous equations

determinant of the *Jacobian* of the system

⇒ it can be employed iteratively to home in on the roots of two simultaneous equations.

Newton-Raphson

Input: a continuous function u, v

1. Evaluate $u'(x,y)$, $v'(x,y)$ symbolically

2. Use an initial guess of the root x_i, y_i to estimate the new value of the root x_{i+1}, y_{i+1} as

3. Find the relative approximate error

Case study 1

Use the multiple-equation Newton-Raphson method to determine roots of equations

$$u(x, y) = x^2 + xy - 10 = 0$$
$$v(x, y) = y + 3xy^2 - 57 = 0$$

Case study 1

$$u(x, y) = x^2 + xy - 10 = 0$$
$$v(x, y) = y + 3xy^2 - 57 = 0$$

Step 1: Initiate the computation with guesses

Step 2: The first derivatives of the function can be evaluated

Case study 1

$$u(x, y) = x^2 + xy - 10 = 0$$
$$v(x, y) = y + 3xy^2 - 57 = 0$$

Step 3: the determinant of the Jacobian for the first iteration

Step 4: The values of the functions can be evaluated at the initial guesses

Case study 1

$$u(x, y) = x^2 + xy - 10 = 0$$

$$v(x, y) = y + 3xy^2 - 57 = 0$$

Step 5: These values can be substituted into the two-equation Newton-Raphson

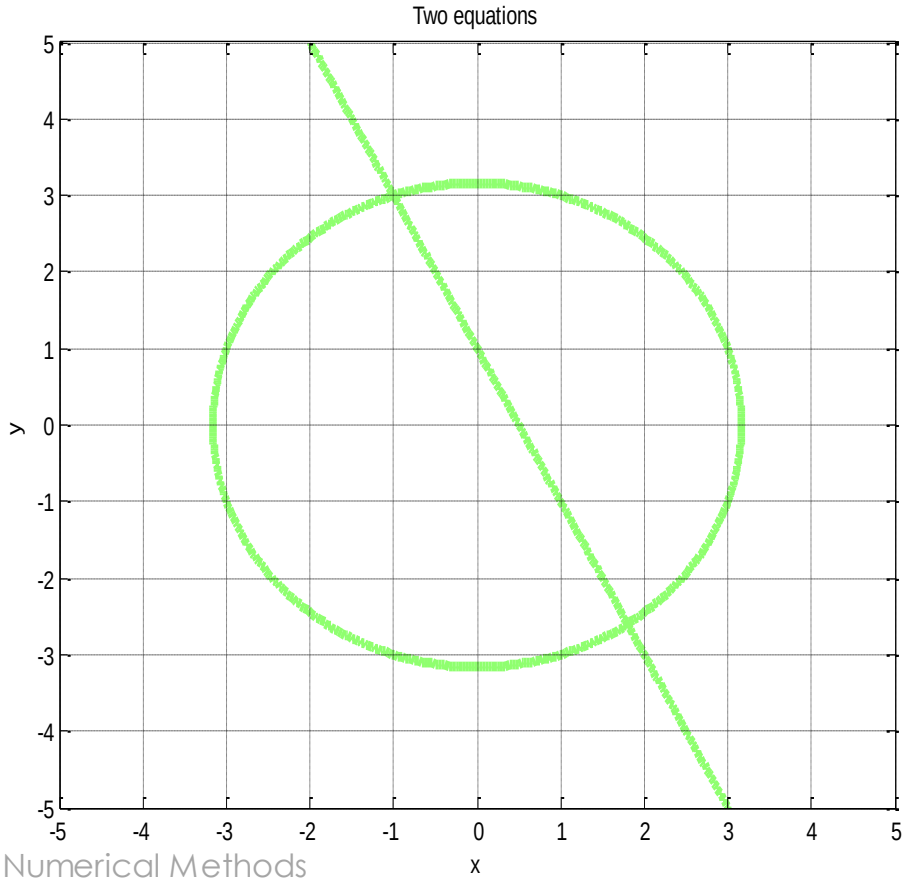
Step 6: the results are converging to the true values of x = 2 and y = 3. The computation can be repeated until an acceptable accuracy is obtained

Case study 2

Use the multiple-equation Newton-Raphson method to determine roots of equations

$$\begin{cases} x^2 + y^2 = 10 \\ 2x + y = 1 \end{cases}$$

$X_0 = 2$
$Y_0 = -2$



Exercise

1. Use the Newton-Raphson method to determine roots of equations

6.3 Determine the highest real root of $f(x) = x^3 - 6x^2 + 11x - 6.1$:

3 iterations, initial guess: $x_1 = 3.5$

2. Use the multiple-equation Newton-Raphson method to determine roots of equations

$$x_1 - x_2 + 1 = 0$$
$$x_1^2 + x_2^2 - 4 = 0.$$

3 iterations, initial guess:
$x_1 = 0.8$
$x_2 = 1.8$

Exercise

2. Use the multiple-equation Newton-Raphson method to determine roots of equations

$$x_1 - x_2 + 1 = 0$$
$$x_1^2 + x_2^2 - 4 = 0.$$

2 iterations, initial guess:
$x_1 = 0.8$
$x_1 = 1.8$

Repeating this proces once more we find the final iterate

$$\mathbf{x}_3 = \begin{bmatrix} 0.82287565553230 \\ 1.82287565553230 \end{bmatrix}$$

which is correct to 16 decimal digits as you may check from the exact solution

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \mathbf{r} = \begin{bmatrix} \frac{1}{2}(\sqrt{7} - 1) \\ \frac{1}{2}(\sqrt{7} + 1) \end{bmatrix}.$$

Matlab

Matlab

```
> C = rand(5,5) .random matrix with uniform distribution in [0, 1].
> size(C) .....gives the dimensions (m × n) of C.
> det(C) ..... the determinant of the matrix.
> max(C) .....the maximum of each column.
> min(C) ..... the minimum in each column.
> sum(C) ..... sums each column.
> mean(C) ..... the average of each column.
> diag(C) .....just the diagonal elements.
> C' ..... tranpose the matrix.
```

In addition to ones, eye, zeros, rand and randn

Exercise

```
%% bisection
f=@(x)(x^3-x-2)
a=1
b=2
[p,i] = bisection(f,a,b)
```

Write the function: bisection(f,a,b) to solve the

Matlab

```
clc
clear all
close all
%%
```

Matlab

```
clc
clear all
close all
%%
```

```
function [ x ] = bisection(f,x_l,x_u,esp_s)
```

```
end
```

Find the root of a non-linear equation `f=@(x)(x^3-x-2)`

The fzero function is designed to find the real root of a single equation.
A simple representation of its syntax is fzero(function,x_0) or fzero(function,[x_0 x_1])

Step 1: To find a zero of the function $f(x) = x^3 - x - 2$, write an anonymous function

Step 2: Then find the zero near one guess

Case study 1-Matlab

Find the eigenvalue of a matrix $A = \begin{bmatrix} 3 & 6 & -8 \\ 0 & 0 & 6 \\ 0 & 0 & 2 \end{bmatrix}$

Step 1: We need to solve the equation

$$\det(A - \lambda I) = 0.$$

Step 2: Calculation

Step 3: calculate the determinant

Case study 2 - Matlab

Find the roots of a polynomial

$$f(x) = x^3 - 7x + 6$$

Step 1: the companion matrix

Step 2: The eigenvalues of A

Step 3: The eigenvalues of A are the polynomial roots

Case study 2 - Matlab

Find the roots of a polynomial

$$f(x) = x^3 - 7x + 6$$

```
%
u = [1   0   -7  6]
n= 3
A = diag(ones(n-1,1),-1);
A(1,:) = -u(2:n+1)/u(1);
x = eig(A)
%
```

```
A =

     0     7    -6
     1     0     0
     0     1     0
```

Case study 2 - Matlab

Find the roots of a polynomial $f(x) = x^3 - 7x + 6$

2nd method:

$$c = [1 \quad 0 \quad -7 \quad 6]$$

$$A = compan(c)$$

$$\Rightarrow \quad A = \begin{bmatrix} 0 & 7 & -6 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

Case study 2 - Matlab

3rd method:

Step 1: coefficients of polynomial

$$syms\ x$$
$$f = x^3 - 7x + 6$$
$$c = \text{sym2poly}(f)$$

Step 2: the companion matrix

$$A = compan(c)$$

$$\Rightarrow \quad A = \begin{bmatrix} 0 & 7 & -6 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

Case study 2 - Matlab

Find the roots of a polynomial

$$f(x) = x^3 - 7x + 6$$

```
clc
clear all
close all
%% CALCULATE THE ROOT OF POLYNOMINALS F = X^3 - 7X + 6
u = [1  0  -7 6]
n = 3
A = diag(ones(n-1,1),-1);
A(1,:) = -u(2:n+1)/u(1);
x = eig(A)
```

Case study 2 - Matlab

Find the roots of a polynomial

$$f(x) = x^3 - 7x + 6$$

```
clc
clear all
close all
%% CALCULATE THE ROOT OF POLYNOMINALS F = X^3 - 7X + 6

% 2nd method
c = [1  0  -7 6]
A = compan(c)
x = eig(A)
```

Case study 2 - Matlab

Find the roots of a polynomial

$$f(x) = x^3 - 7x + 6$$

```
clc
clear all
close all
%% CALCULATE THE ROOT OF POLYNOMINALS F = X^3 - 7X + 6

% 3rd method
syms x
f = x^3 -7*x + 6
c = sym2poly(f)
A = compan(c)
x = eig(A)
```

Case study 2 - Matlab

Find the roots of a polynomial

$$f(x) = x^3 - 7x + 6$$

```
clc
clear all
close all
%% CALCULATE THE ROOT OF POLYNOMINALS F = X^3 - 7X + 6

% 4th method
syms x
f = x^3 -7*x + 6
c = sym2poly(f)
x = roots(c)

p = poly(x)
```

Case study 3 - Matlab

Find the roots of a system

$$u(x, y) = x^2 + xy - 10 = 0$$

$$v(x, y) = y + 3xy^2 - 57 = 0$$

```
%%   Solver of non linear systems:
x=[1.5  3.5]'; % initial guess
N = 5000;
n = 2;
for k=1:N,
    F=[ x(1)*x(1)+ x(1)*x(2) - 10 ;  x(2) + 3*x(1)*x(2)*x(2)-57];
    J=[2*x(1)+x(2), x(1) ; 3*x(2).^2 , 1 + 6*x(1).*x(2)];
    dx=J\F;
    x=x-dx
    k=k+1
    pause
end
x'
```

Case study 3 - Matlab

Find the roots of a system

$$u(x, y) = x^2 + xy - 10 = 0$$
$$v(x, y) = y + 3xy^2 - 57 = 0$$

```
clc
clear all
close all

fun = @myfun;
x0 = [0,0];
options=optimset('Display','iter');
%Option to display output
[x,fval] = fsolve(fun,x0,options)
```

```
function F = myfun(x)

F(1) = x(1)^2 + x(1)*x(2) - 10;
F(2) = x(2) +3*x(1)*x(2)^2 -57 ;
```